5-4 THE PAIRED *t*-TEST

- **Definition**: A paired t-test is a special case of the two-sample t-test used when observations are collected in **pairs** under homogeneous conditions.

- **Controlled Conditions**: While conditions within each of the **pairs** are consistent, they may vary significantly from one pair to another.

- **Experimental Design**: For example, when testing different hardness-testing tips, using independent samples could lead to errors because the specimens themselves (the bar stock) might differ in hardness.

- **Error Prevention**: By using **pairs**, the test ensures that the observed differences are due to the treatment (the tips) rather than the inherent variability between different specimens.

- **Experimental Procedure**: A more effective method involves collecting data in pairs by taking two readings on a single specimen, then analyzing the **differences** between those readings.
- **Null Hypothesis**: If the two tools being tested are identical, the average of these **differences** should be zero.
- **Statistical Method**: This specific approach to analyzing related data points is known as the **paired t-test**.

- **Data Structure**: The procedure uses n paired observations, (X_{1j}, X_{2j}), where each population has its own mean (μ_1, μ_2) and variance (σ_1^2, σ_2^2).

- **Variable Definition**: The differences are defined as $D_j = X_{1j} - X_{2j}$. These differences are assumed to be normally distributed.

- **Parameter Relationship**: The mean of the differences, μ_D, is equal to the difference between the two population means $(\mu_1 - \mu_2)$.

- **Hypothesis Equivalence**: Testing the difference between two population means is simplified into a one-sample t-test on μ_D. The hypotheses are defined as:

$$H_0 : \mu_D = \Delta_0 \qquad (5-15)$$
$$H_1 : \mu_D \neq \Delta_0$$

- **The Paired t-Test**: This statistical procedure is used to evaluate the null hypothesis H_0: $\mu_D = \Delta_0$ for observations collected in pairs.

- **Test Statistic**: The calculation is centered as follows, where s_d represents the sample standard deviation or **standard error**:

$$t_0 = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}} \qquad (5-16)$$

- **Testing Criteria**: The framework defines the **Alternative Hypothesis**, **P-Value**, and **Rejection Criterion for Fixed-Level Tests** based on the direction of the test:
  - **Two-sided**:
    - **Alternative Hypothesis**: $H_1: \mu_D \neq \Delta_0$
    - **P-Value**: $2[1 - F_{t_{n-1}}(|t_0|)]$
    - **Rejection Criterion for Fixed-Level Tests**: $|t_0| > t_{\alpha/2,n-1}$

- Upper-tailed:
  - Alternative Hypothesis: H_1: $\mu_D > \Delta_0$
  - P-Value: $1 - F_{t_{n-1}}(t_0)$
  - Rejection Criterion for Fixed-Level Tests: $t_0 > t_{\alpha,n-1}$
- Lower-tailed:
  - Alternative Hypothesis: H_1: $\mu_D < \Delta_0$
  - P-Value: $F_{t_{n-1}}(t_0)$
  - Rejection Criterion for Fixed-Level Tests: $t_0 < -t_{\alpha,n-1}$

- **Definitions in (5-16)**: $\bar{d}$ is the sample average of the n differences, while s_d is the **standard error** or sample standard deviation of those differences.

Case 3: Paired t-Test

- **Null Hypothesis:** H_0: $\mu_D = \Delta_0$
- **Test Statistic**: Based on the sample mean ($\bar{d}$) and sample standard deviation (s_d) of the differences, scaled by its estimated standard error ($s_d/\sqrt{n}$):

$$t_0 = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}} \quad \textbf{(5-16)}$$

- **Differences and Degrees of Freedom:**
  - For each pair j, the difference is: $d_j = x_{1j} - x_{2j}$
  - The number of degrees of freedom is: $df = n - 1$

- **Confidence Interval:** A $100(1-\alpha)\%$ confidence interval on the mean difference μ_D is:

$$\bar{d} \pm t_{\alpha/2,n-1} \cdot \frac{s_d}{\sqrt{n}} \quad (\mathbf{5-17})$$

where $t_{\alpha/2,n-1}$ is the upper $100\alpha/2$ percentage point of the t distribution.

- **Decision Criteria**: The table below details the **Alternative Hypotheses**, **P-Value** calculations, and the **Rejection Criterion for Fixed-Level Tests:**

Alternative Hypotheses	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: \mu_D \neq \Delta_0$	$2[1 - F_{t_{n-1}}(\|t_0\|)]$	$\|t_0\| > t_{\alpha/2,n-1}$
$H_1: \mu_D > \Delta_0$	$1 - F_{t_{n-1}}(t_0)$	$t_0 > t_{\alpha,n-1}$
$H_1: \mu_D < \Delta_0$	$F_{t_{n-1}}(t_0)$	$t_0 < -t_{\alpha,n-1}$

- **EXAMPLE 5-8 Shear Strength of Steel Girders**: This study, published in the **Journal of Strain Analysis**, compares two methods for predicting the shear strength of steel plate girders.

- **Table 5-3**: The following data represents the predicted-to-observed load ratios for nine girders:

Table 5-3 Strength Predictions for Nine Steel Plate Girders (Predicted Load/Observed Load)

Girder	Karlsruhe Method	Lehigh Method	Difference d_j
S1/1	1.186	1.061	0.125
S2/1	1.151	0.992	0.159
S3/1	1.322	1.063	0.259
S4/1	1.339	1.062	0.277
S5/1	1.200	1.065	0.135
S2/1	1.402	1.178	0.224
S2/2	1.365	1.037	0.328
S2/3	1.537	1.086	0.451
S2/4	1.559	1.052	0.507

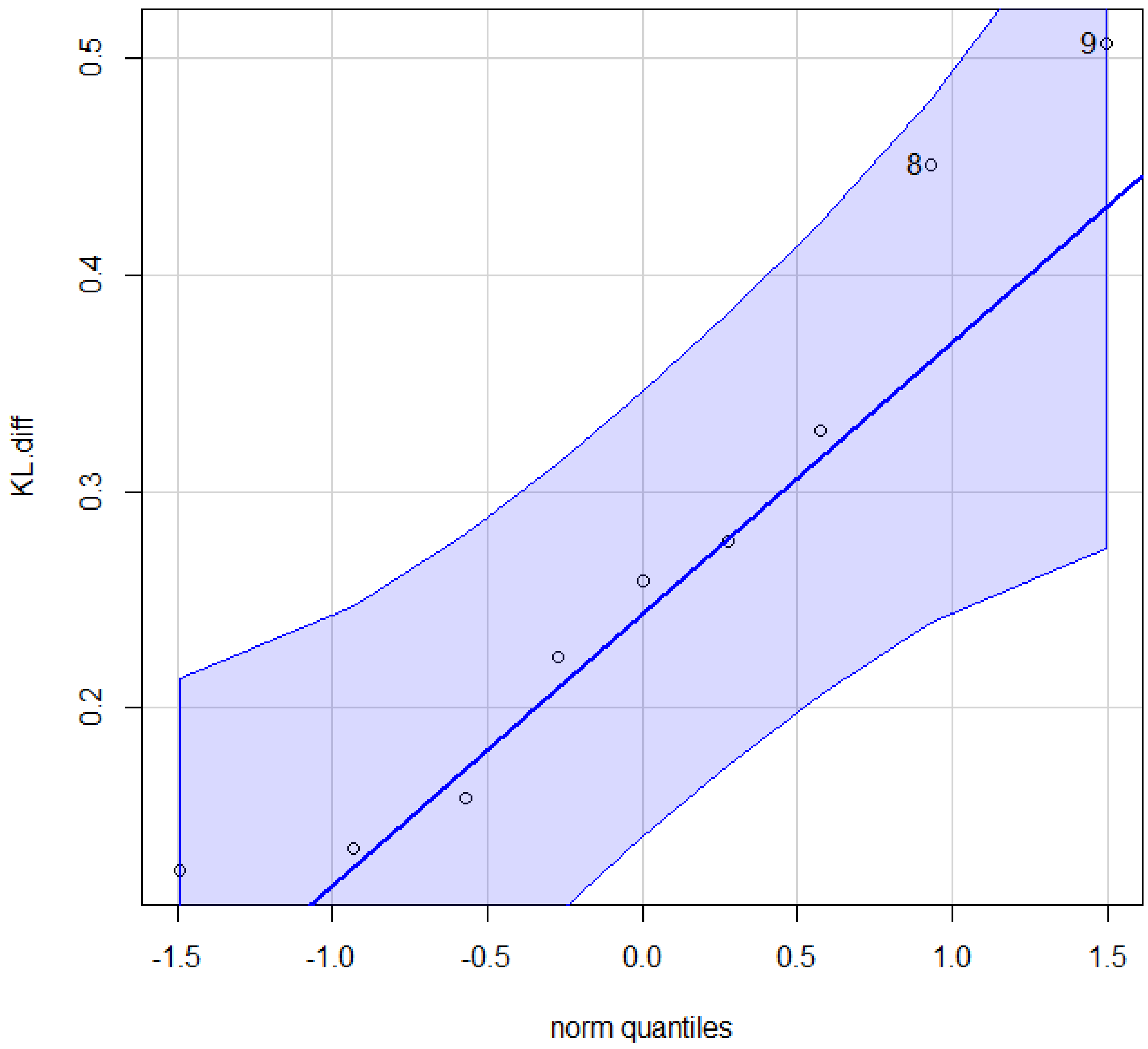
Solution.

- **1. Parameter of interest**: The difference in mean shear strength between the two methods, $\mu_D = \mu_1 - \mu_2$.
- **2. Null hypothesis, H_0** : $\mu_D = 0$.
- **3. Alternative hypothesis, H_1** : $\mu_D \neq 0$.
- **4. Test statistic**: The formula $t_0 = \bar{d}/(s_d/\sqrt{n})$.
- **5. Reject H_0 if**: The p-value is < 0.05.

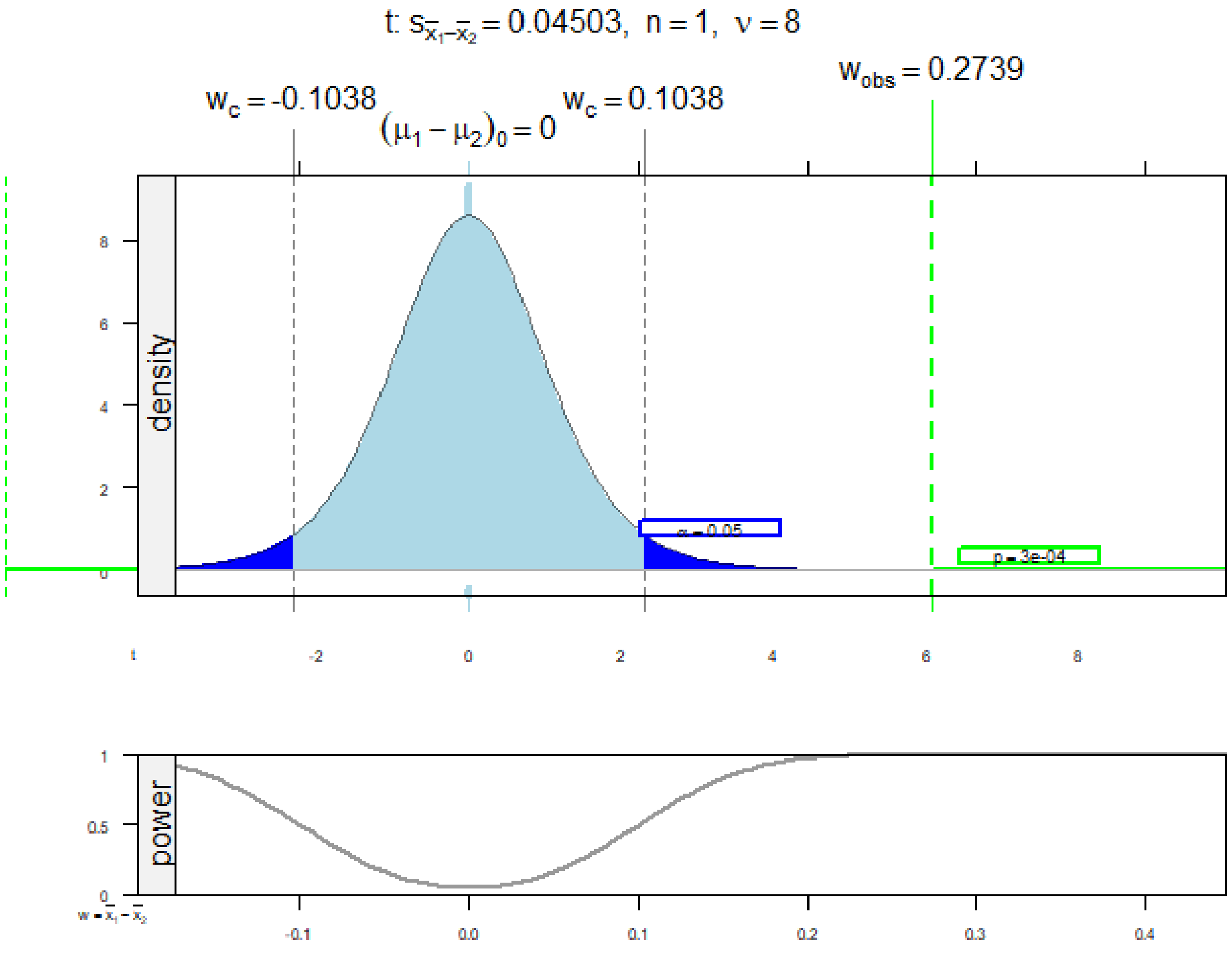
- **6. Computations**: With $\bar{d} = 0.2738889$ and $s_d = 0.1350994$, the resulting test statistic is $t_0 = 6.081939$.

- **7. Conclusions**: The p-value is approximately 0.0003. Since this is less than 0.05, we conclude the methods yield different results.

- **Practical engineering conclusion:** The data indicate the Karlsruhe method produces higher average strength predictions than the Lehigh method.



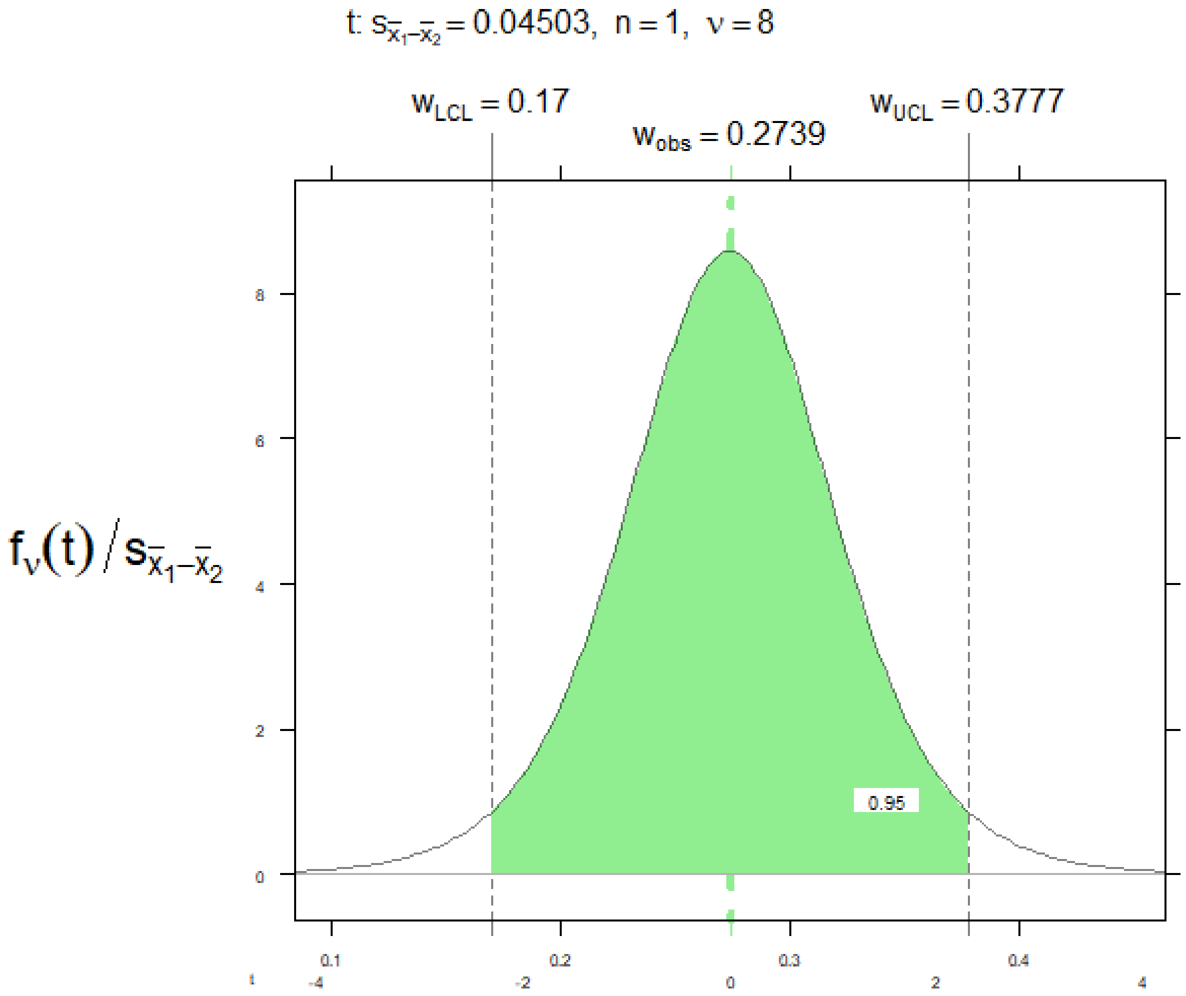
KL.diff
norm quantiles
0.2
0.3
0.4
0.5
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
8
9



Paired t-test: Karlsruhe and Lehigh

	$(\mu_1-\mu_2)_0$	w_{obs}	w_{other}	w_{critL}	w_{critR}
$\bar{x}$	0.000	0.274	-0.274	-0.104	0.104
t	0.000	6.082	-6.082	-2.306	2.306

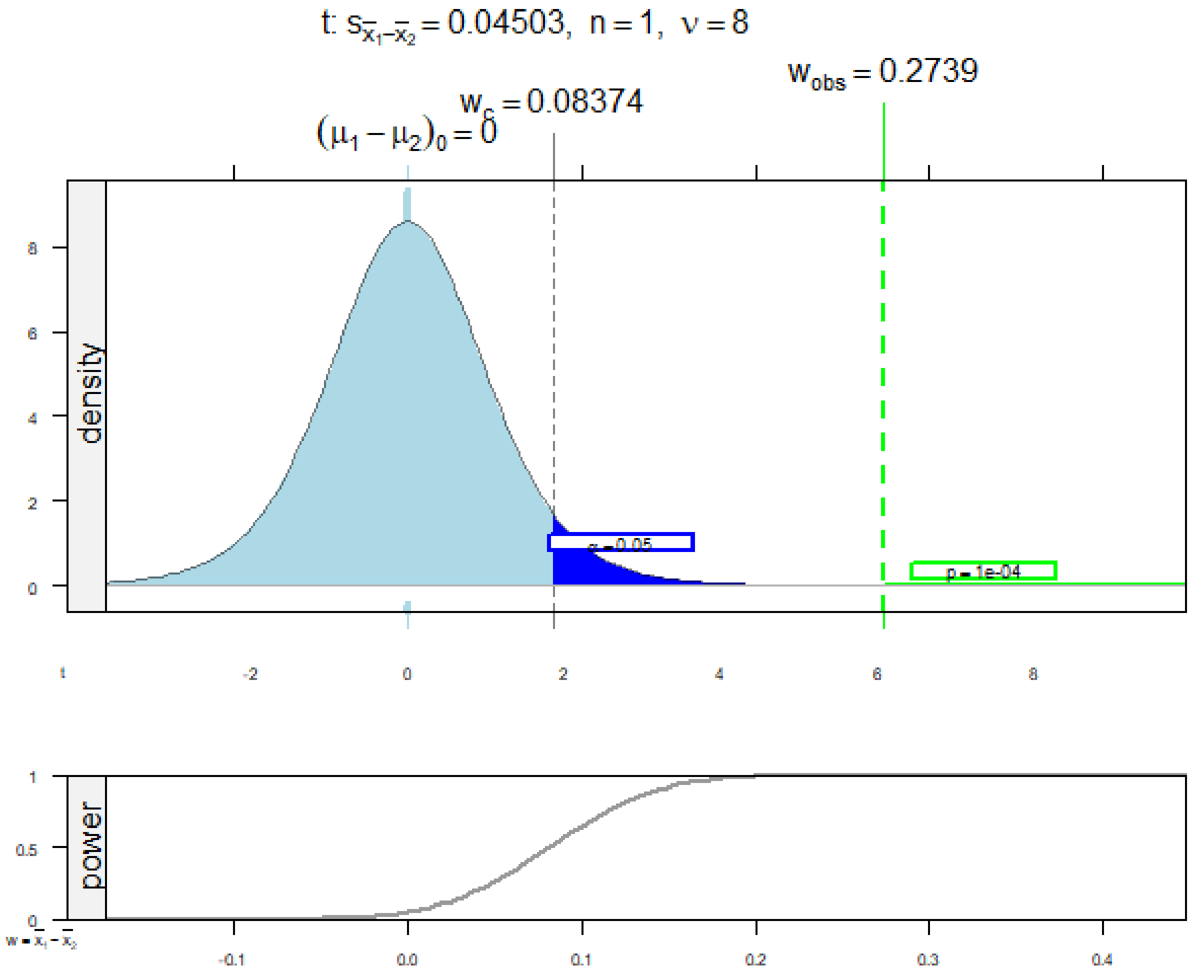
	Left	Combined	Right
p	0.0001	0.0003	0.0001
α	0.0250	0.0500	0.0250



Paired t-test: Karlsruhe and Lehigh

	w_{obs}	w_{LCL}	w_{UCL}
$\bar{x}$	0.2739	0.1700	0.3777
t	0.0000	-2.3060	2.3060

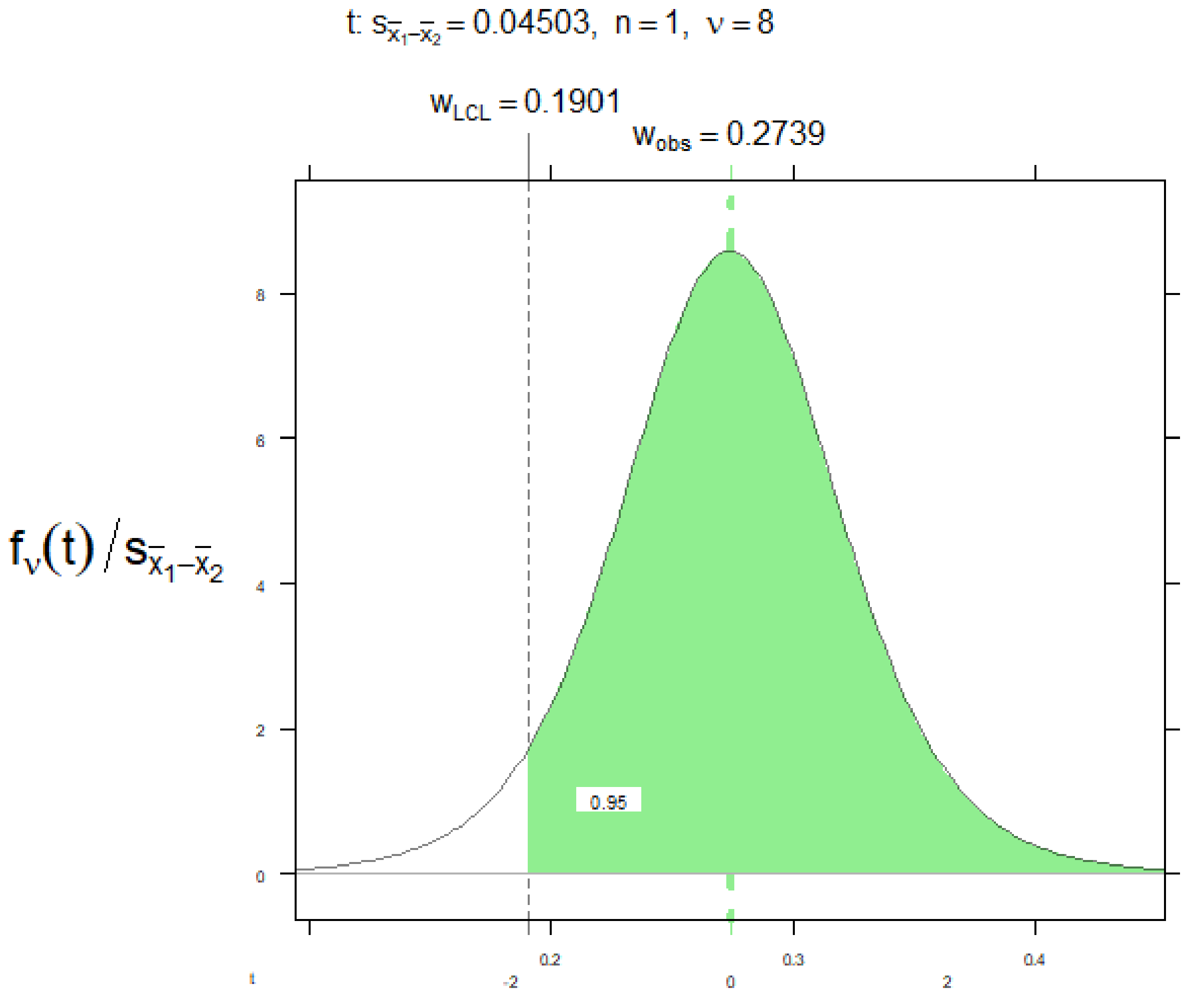
	Left	Confidence	Right
Probability	0.0250	0.9500	0.0250



Paired t-test: Karlsruhe and Lehigh

	$(\mu_1-\mu_2)_0$	w_{obs}	w_{critLR}
$\bar{x}$	0.000	0.274	0.084
t	0.000	6.082	1.860

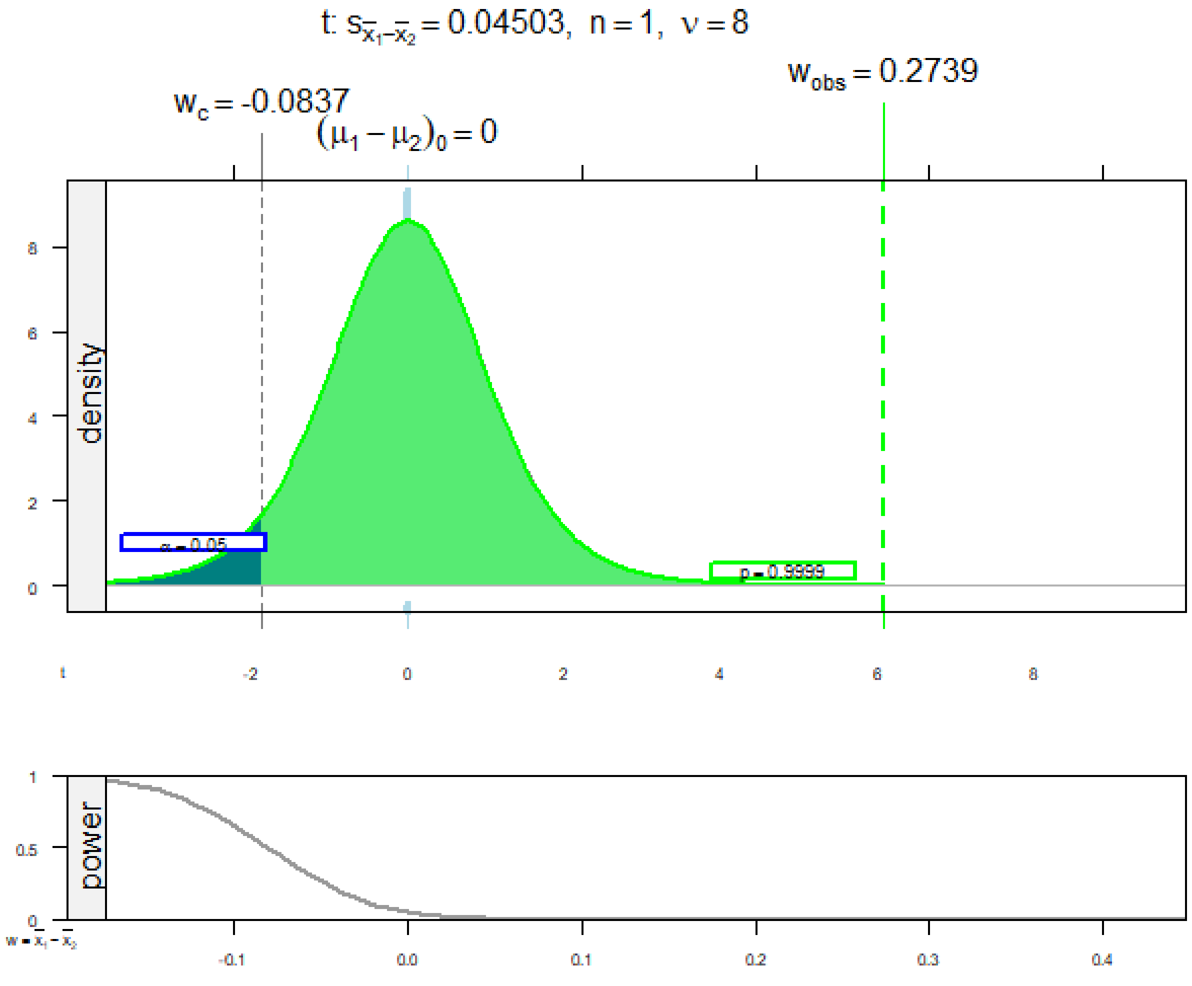
	Probability
p	1e-04
α	5e-02



Paired t-test: Karlsruhe and Lehigh

	w_{obs}	w_{LCL}
$\bar{x}$	0.2739	0.1901
t	0.0000	-1.8595

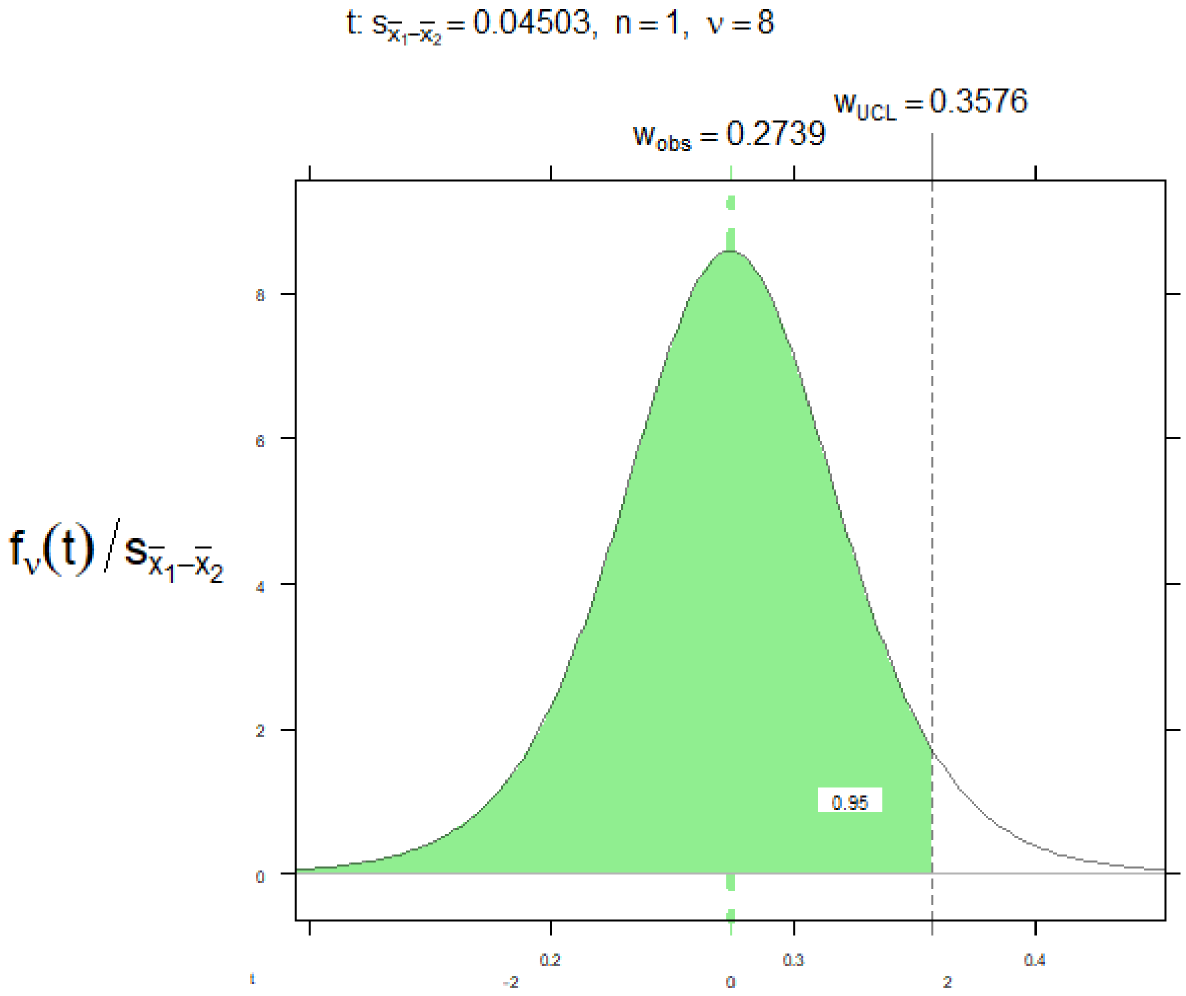
	Probability
Confidence	0.9500



Paired t-test: Karlsruhe and Lehigh

	$(\mu_1-\mu_2)_0$	w_{obs}	$w_{crit.L}$
$\bar{x}$	0.000	0.274	-0.084
t	0.000	6.082	-1.860

	Probability
p	0.9999
α	0.0500



Paired t-test: Karlsruhe and Lehigh

	w_{obs}	w_{UCL}
$\bar{x}$	0.2739	0.3576
t	0.0000	1.8595

	Probability
Confidence	0.9500

Checklist for Selecting a t-Test

1. How many groups are being compared?

- **One Group**: Use a **one-sample t-test** to compare the group mean to a known standard or theoretical value.
- **Two Groups**: Proceed to the next question. GraphPad +3

2. Are the groups independent or dependent?

- **Dependent (Related)**: Use **Case 3: Paired t-Test**. This applies if you are measuring the same subjects twice (e.g., before and after treatment) or using matched pairs.
- **Independent**: The groups consist of different people or items (e.g., Treatment A vs. Treatment B). Proceed to the final question. YouTube +3

3. Are the population variances equal?

- **Equal Variances (Assume $\sigma_1^2 = \sigma_2^2$)**: Use **Case 1: Pooled t-Test**. This is typically verified using an **F-test**.
- **Unequal Variances (Assume $\sigma_1^2 \neq \sigma_2^2$)**: Use **Case 2: Welch t-test**. This is the most robust choice if you are unsure or if sample sizes are very different. Educational Research Basics by Del Si... +2

- **Confidence Interval for μ_D**: The **Confidence Interval on μ_D for Paired Observations** is derived using the fact that the ratio $T = \bar{D}/(S_D/\sqrt{n})$ follows a t-distribution with $n-1$ degrees of freedom.

- **Probability Basis**: By setting the probability $P(-t_{\alpha/2,n-1} \le T \le t_{\alpha/2,n-1}) = 1-\alpha$, we can isolate the mean difference μ_D to establish the upper and lower bounds.

- **Confidence Interval Formula**: For a $100(1-\alpha)\%$ confidence level, the interval is defined as:

$$\bar{d} - t_{\alpha/2,n-1} \cdot s_d/\sqrt{n} \le \mu_D \le \bar{d} + t_{\alpha/2,n-1} \cdot s_d/\sqrt{n} \qquad (5-17)$$

- **Components of (5-17):**
  - $\bar{d}$: Sample mean of the differences.
  - s_d: Sample standard deviation of the differences.
  - $t_{\alpha/2,n-1}$: The upper $100\alpha/2$ percentage point of the t-distribution with $n-1$ degrees of freedom.

- **EXAMPLE 5-9 Parallel Parking**: This study, published in the journal **Human Factors**, investigates the time required for $n = 14$ subjects to parallel park two different automobiles with distinct wheelbases and turning radii.

- **Data Collection**: The performance of each subject was recorded in seconds, focusing on the paired difference (d_j) for each individual.

Table 5-4 Time in Seconds to Parallel Park Two Automobiles

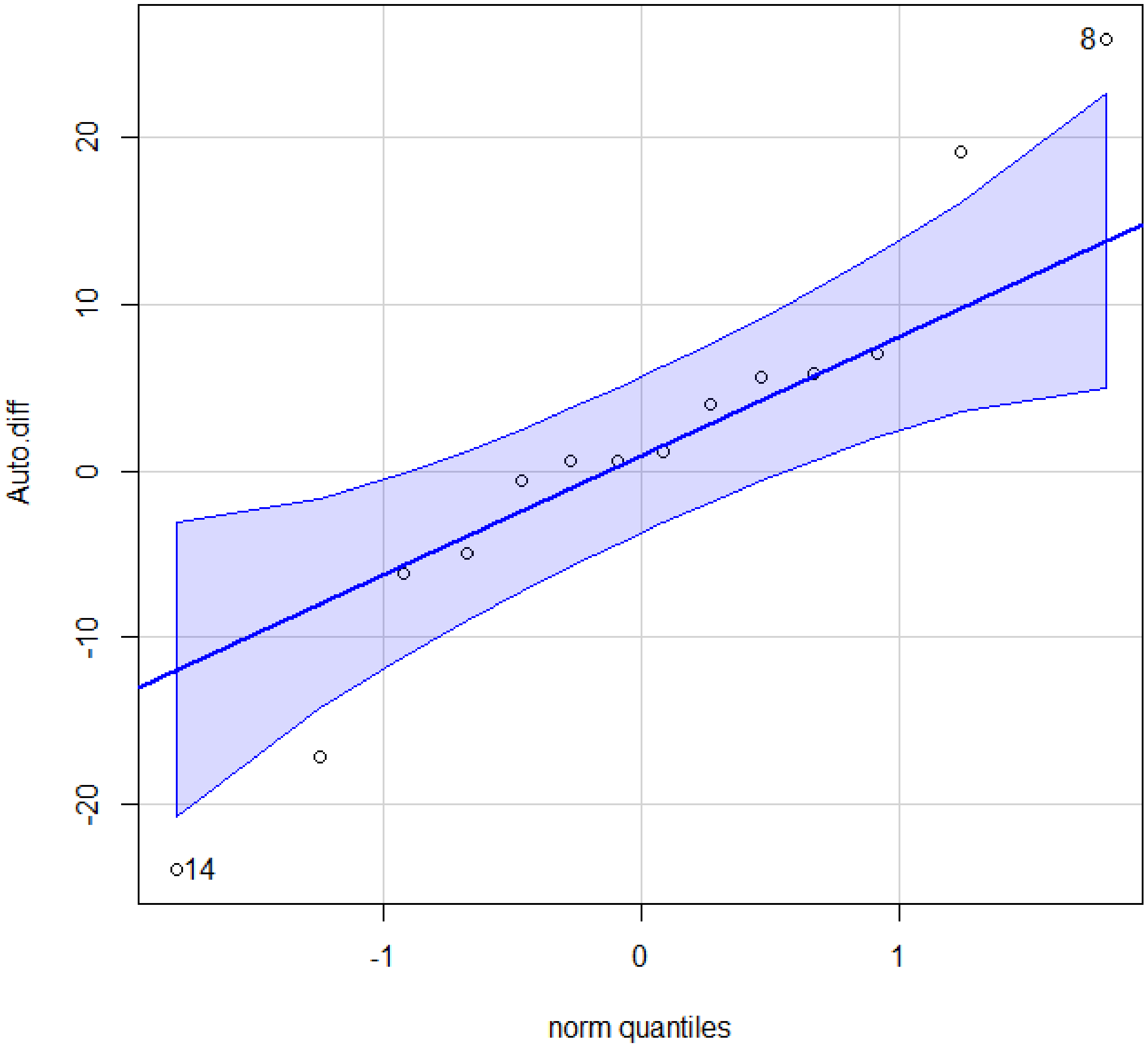
Subject	Automobile 1 (x_{1j})	Automobile 2 (x_{2j})	Difference (d_j)
1	37.0	17.8	19.2
2	25.8	20.2	5.6
3	16.2	16.8	-0.6
4	24.2	41.4	-17.2
5	22.0	21.4	0.6
6	33.4	38.4	-5.0
7	23.8	16.8	7.0
8	58.2	32.2	26.0
9	33.6	27.8	5.8
10	24.4	23.2	1.2
11	23.4	29.6	-6.2
12	21.2	20.6	0.6
13	36.2	32.2	4.0
14	29.8	53.8	-24.0

Solution.

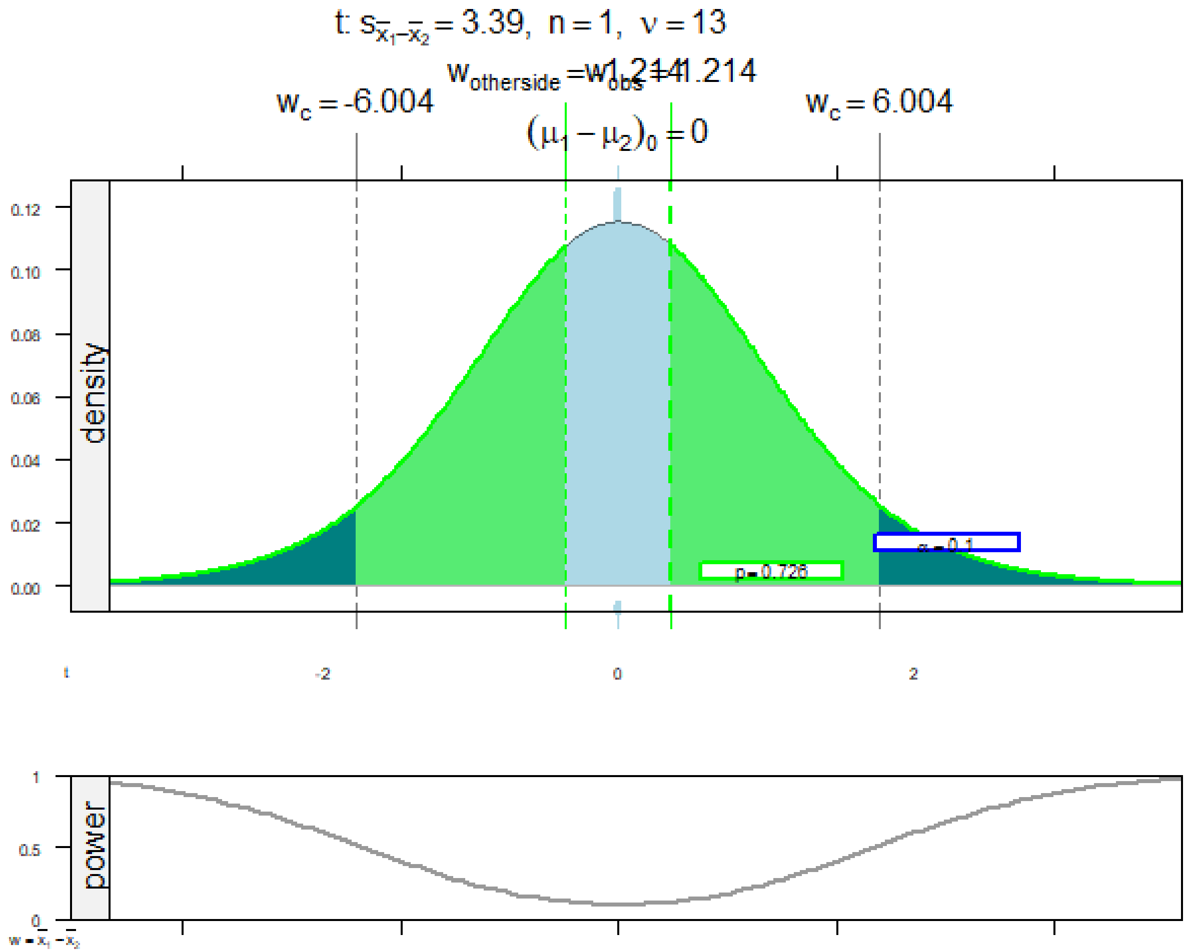
- **Statistical Parameters**: Based on the differences, the sample mean difference is $\bar{d} = 1.214$ and the standard deviation is $s_d = 12.685$.
- **Confidence Interval Calculation**: Using a t-distribution with 13 degrees of freedom and a 90% confidence level ($t_{0.05,13} = 1.771$):

$$-4.79 \leq \mu_D \leq 7.22$$

- **Conclusion**: Because the confidence interval includes zero, the data do not support the claim that there is a significant difference in mean parking times between the two cars at the 90% confidence level.



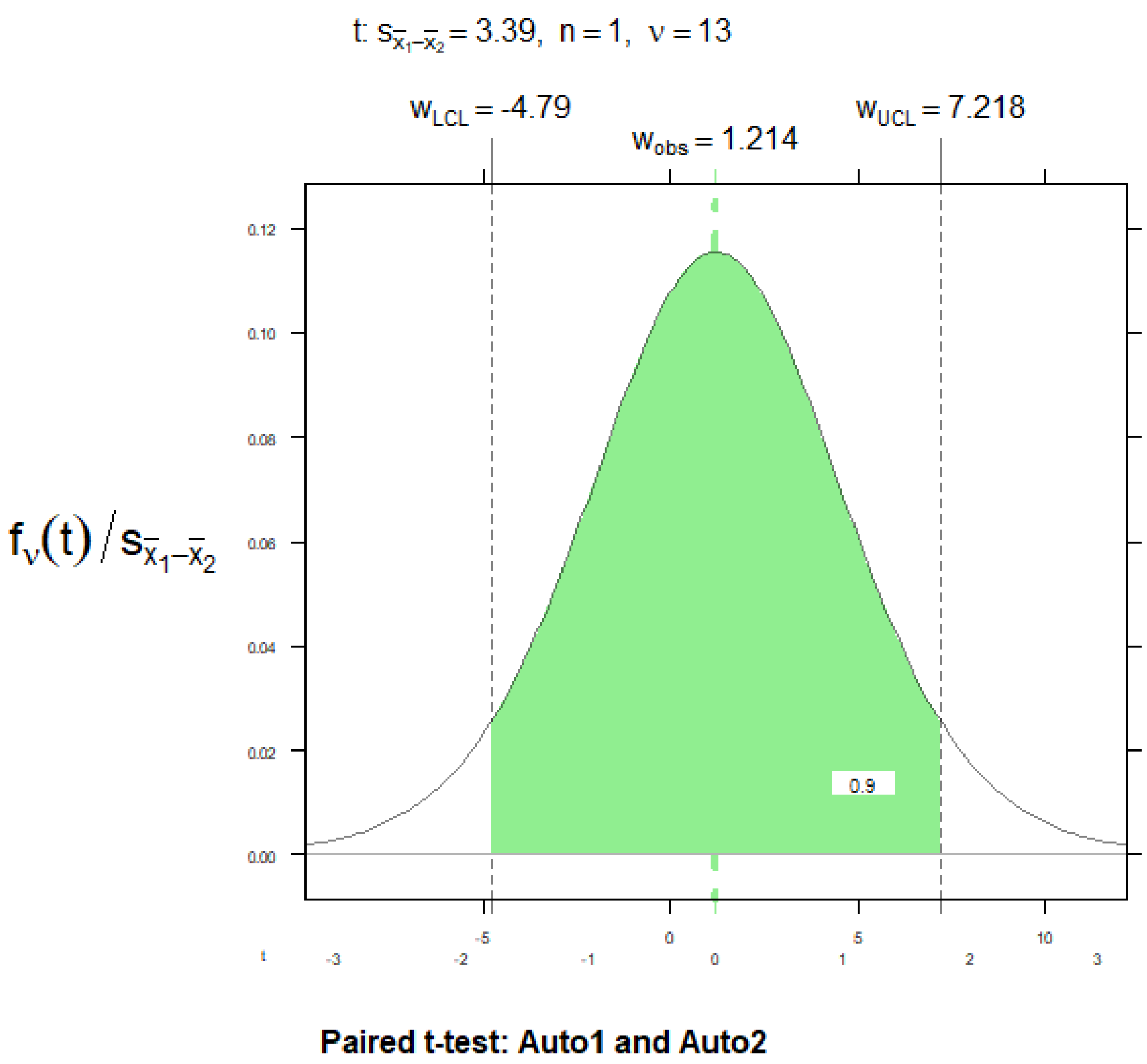
8
14
20
10
0
-10
-20
Auto.diff
-1
0
1
norm quantiles



Paired t-test: Auto1 and Auto2

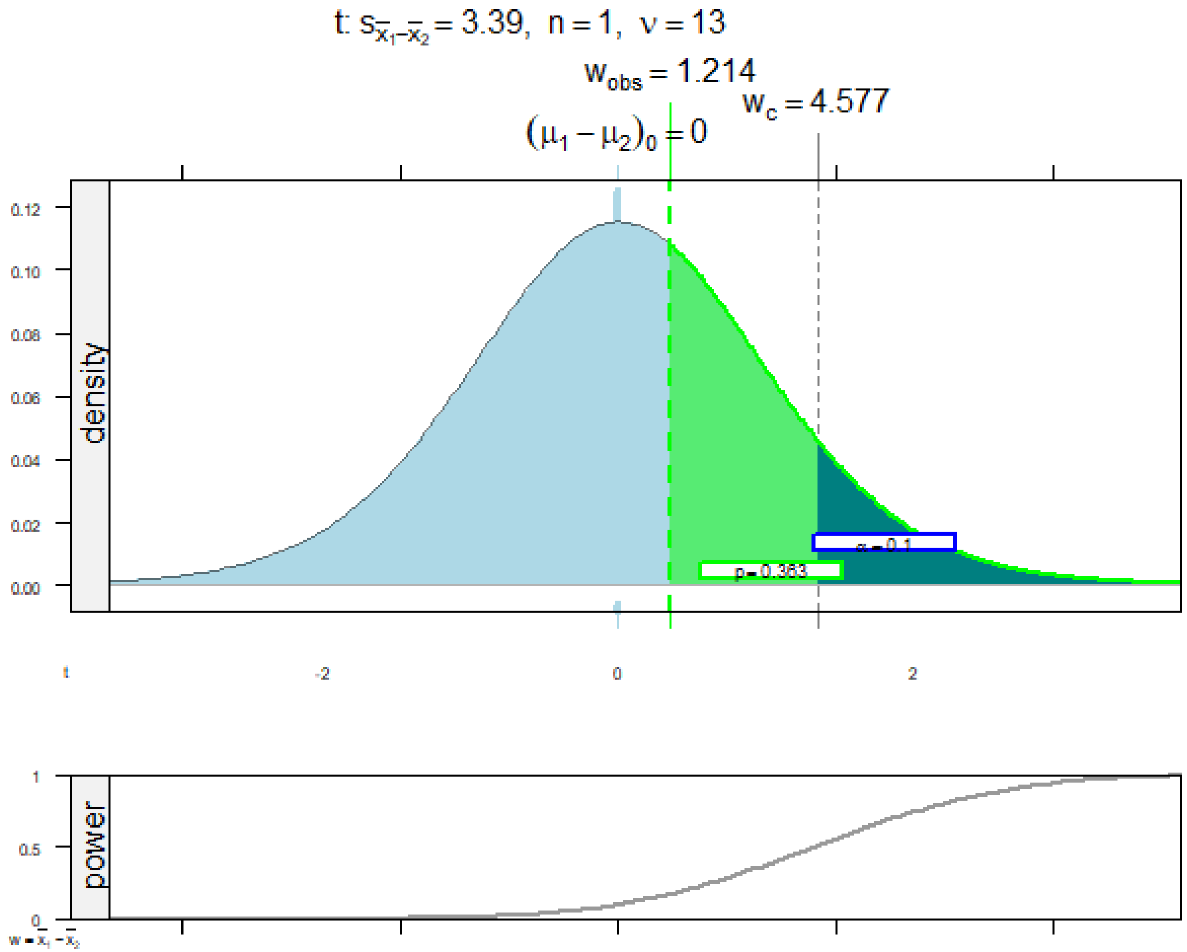
	$(\mu_1-\mu_2)_0$	w_{obs}	w_{other}	w_{critL}	w_{critR}
$\bar{x}$	0.000	1.214	-1.214	-6.004	6.004
t	0.000	0.358	-0.358	-1.771	1.771

	Left	Combined	Right
p	0.3630	0.7260	0.3630
α	0.0500	0.1000	0.0500



	w_{obs}	w_{LCL}	w_{UCL}
$\bar{x}$	1.214	-4.790	7.218
t	0.000	-1.771	1.771

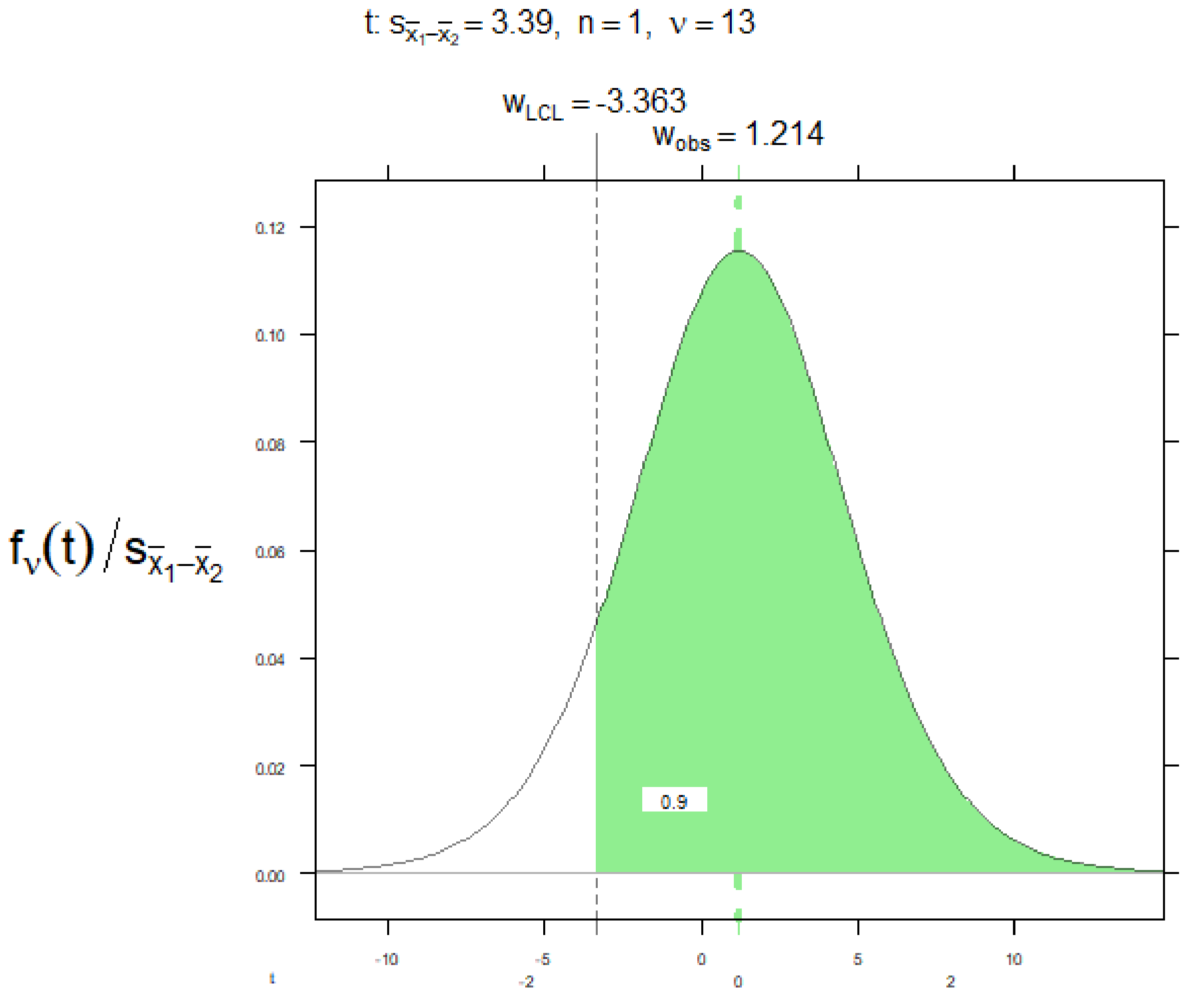
	Left	Confidence	Right
Probability	0.0500	0.9000	0.0500



Paired t-test: Auto1 and Auto2

	$(\mu_1-\mu_2)_0$	w_{obs}	w_{critLR}
$\bar{x}$	0.000	1.214	4.577
t	0.000	0.358	1.350

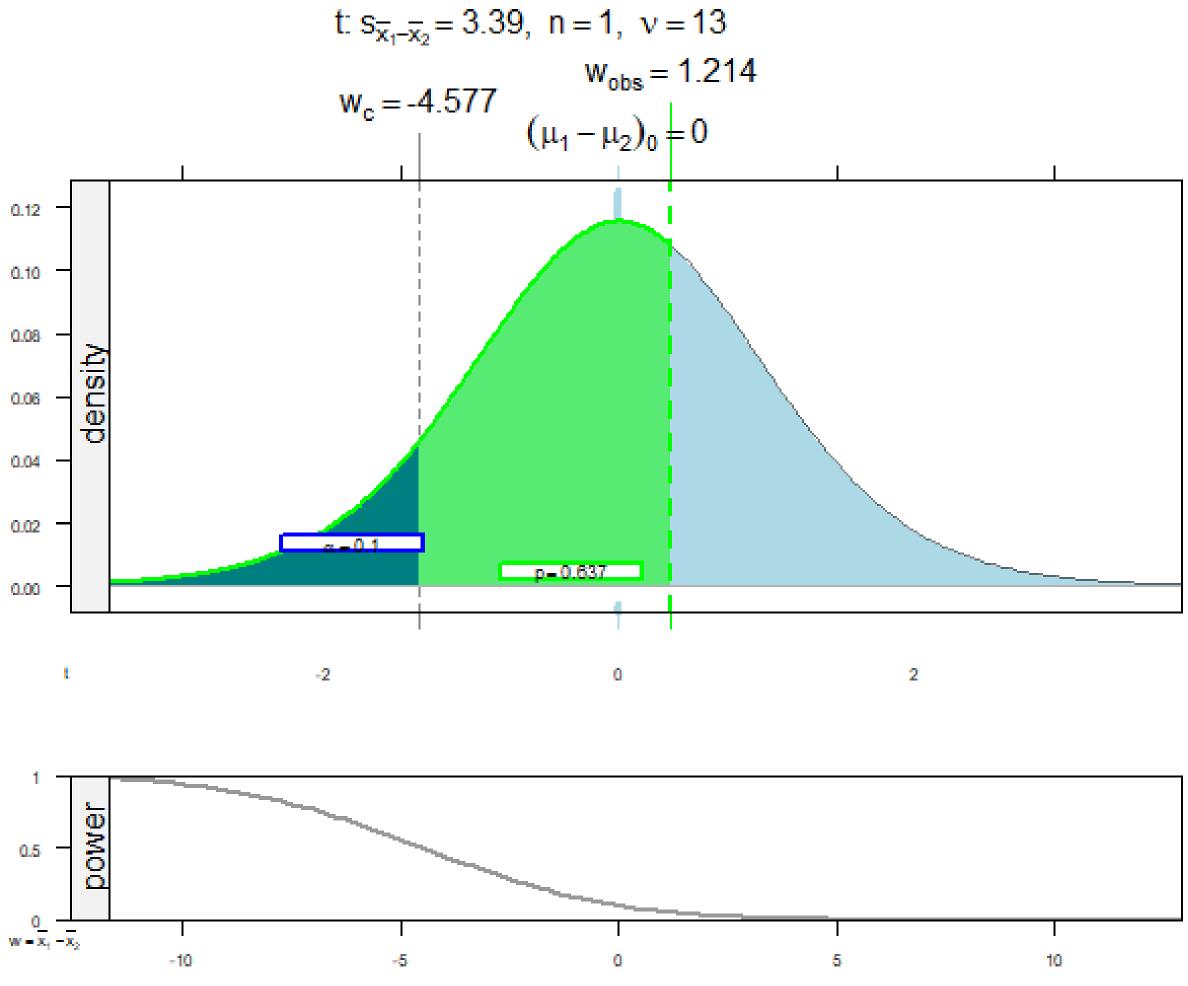
	Probability
p	0.3630
α	0.1000



Paired t-test: Auto1 and Auto2

	w_{obs}	w_{LCL}
$\bar{x}$	1.214	-3.363
t	0.000	-1.350

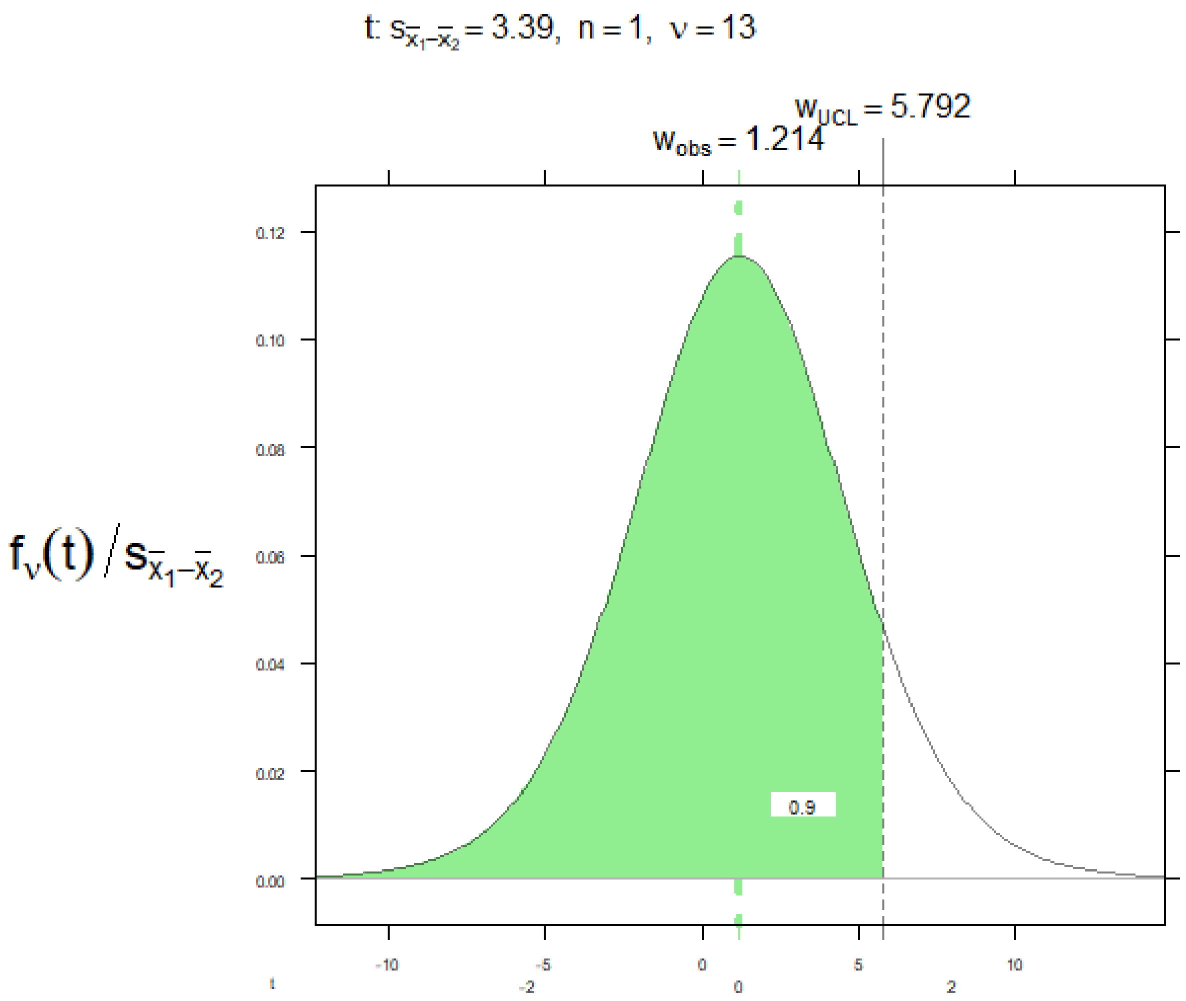
	Probability
Confidence	0.9000



Paired t-test: Auto1 and Auto2

	$(\mu_1-\mu_2)_0$	w_{obs}	w_{critLL}
$\bar{x}$	0.000	1.214	-4.577
t	0.000	0.358	-1.350

	Probability
p	0.6370
α	0.1000



Paired t-test: Auto1 and Auto2

	w_{obs}	w_{UCL}
$\bar{x}$	1.214	5.792
t	0.000	1.350

	Probability
Confidence	0.9000